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exam1@sysadmin-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Downloads/hostel_final$ java BlackBoxTests
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ECP01: [Valid] All valid inputs, rooms available, fee PAID
[NOTIFY] Allotment: Student S001 -> Room 301 Block B
[PASS] ECP01 All valid -> ALLOTTED
ECP02: [Invalid] studentId = null
[PASS] ECP02 null studentId -> InvalidStudentException
ECP03: [Invalid] studentId not registered in system
[PASS] ECP03 Unregistered student -> InvalidStudentException
ECP04: [Invalid] roomPreference = 7 (out of range)
[PASS] ECP04 pref=7 -> InvalidPreferenceException
ECP05: [Valid-edge] No rooms available -> WAITLISTED
[NOTIFY] Waitlist: Student S001 added to waitlist.
[PASS] ECP05 No rooms -> WAITLISTED
ECP06: [Valid-edge] Rooms available but fee = PENDING
[NOTIFY] Allotment: Student S001 -> Room 301 Block B
[NOTIFY] Cancelled: Allotment for student S001 cancelled due to non-payment.
[PASS] ECP06 Fee PENDING -> ALLOTMENT_CANCELLED + room freed
ECP07: [Valid] Two students, both get different rooms
[NOTIFY] Allotment: Student S001 -> Room 301 Block B
[NOTIFY] Allotment: Student S002 -> Room 201 Block A
[PASS] ECP07 Two students -> allotted different rooms
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--- BVA: Boundary Value Analysis ---

Boundaries: roomPreference [1-5], roomCapacity [0,1,2,N]

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BVA01: roomPreference = 0 (below minimum)
[PASS] BVA01 pref=0 -> InvalidPreferenceException
BVA02: roomPreference = 1 (minimum valid)
[NOTIFY] Allotment: Student S001 -> Room 101 Block A
[PASS] BVA02 pref=1 -> ALLOTTED: 101
BVA03: roomPreference = 5 (maximum valid)
[NOTIFY] Allotment: Student S001 -> Room 501 Block C
[PASS] BVA03 pref=5 -> ALLOTTED: 501
BVA04: roomPreference = 6 (above maximum)
[PASS] BVA04 pref=6 -> InvalidPreferenceException
BVA05: roomCapacity = 0 (no rooms in DB)
[NOTIFY] Waitlist: Student S001 added to waitlist.
[PASS] BVA05 0 rooms -> WAITLISTED + added to waitlist
BVA06: roomCapacity = 1 (exactly one room)
[NOTIFY] Allotment: Student S001 -> Room 301 Block B
[PASS] BVA06 1 room -> ALLOTTED: 301
BVA07: roomCapacity = 1, second student should be WAITLISTED
[NOTIFY] Allotment: Student S001 -> Room 301 Block B
[NOTIFY] Waitlist: Student S002 added to waitlist.
[PASS] BVA07 S001 ALLOTTED, S002 WAITLISTED
BVA08: feeStatus = null (missing payment)
[NOTIFY] Allotment: Student S001 -> Room 301 Block B
[NOTIFY] Cancelled: Allotment for student S001 cancelled due to non-payment.
[PASS] BVA08 feeStatus=null -> ALLOTMENT_CANCELLED
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RESULTS: PASSED=15 FAILED=0 TOTAL=15
PASS RATE: 100%
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exam1@sysadmin-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Downloads/hostel_final$
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exam1@sysadmin-HP-ProOne-440-23-8-inch-G9-All-in-One-Desktop-PC:~/Downloads/hostel_final$ java WhiteBoxTests
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N9=D4 N11=rankMerit N12=nullCheck N14=allot  
N15=D5 N17=confirm N18=D6 N20=return ALLOTTED
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WB01: [P1] D1=TRUE (null studentId) -> N1-N2-N3  
[PASS] WB01 null studentId -> InvalidStudentException  
WB02: [P1-alt] D1=TRUE (whitespace studentId) -> N1-N2-N3  
[PASS] WB02 whitespace studentId -> InvalidStudentException  
WB03: [P2] D1=FALSE, D2=TRUE (unregistered student) -> N1-N2-N4-N5  
[PASS] WB03 unregistered student -> InvalidStudentException  
WB04: [P3] D3=TRUE (pref=0, below min) -> N1-N2-N4-N6-N7  
[PASS] WB04 pref=0 -> InvalidPreferenceException  
WB05: [P3-alt] D3=TRUE (pref=6, above max) -> N1-N2-N4-N6-N7  
[PASS] WB05 pref=6 -> InvalidPreferenceException  
WB06: [P4] D4=TRUE (no rooms in D8) -> N1-N9-N10  
[NOTIFY] Waitlist: Student S001 added to waitlist.  
[PASS] WB06 No rooms -> WAITLISTED + added to waitlist  
WB07: [P5] D5=TRUE (fee PENDING) -> N1-N15-N16  
[NOTIFY] Allotment: Student S001 -> Room 301 Block B  
[NOTIFY] Cancelled: Allotment for student S001 cancelled due to non-payment.  
[PASS] WB07 fee PENDING -> ALLOTMENT CANCELLED + room freed  
WB08: [P7] Full happy path - all decisions FALSE -> N1-N20  
[NOTIFY] Allotment: Student S001 -> Room 301 Block B  
[PASS] WB08 Happy path -> ALLOTTED: 301 + confirmed + room OCCUPIED
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--- Additional Branch / Statement / Loop Coverage Tests ---
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WB09: [P6] D6=TRUE (biometric fail) -> N1-N18-N19  
[NOTIFY] Allotment: Student S001 -> Room 301 Block B  
[PASS] WB09 biometric fail -> InvalidStudentException (correct msg)  
WB10: rankByMerit exact-match loop (L112-113) - pref=1 hits Room 101  
[NOTIFY] Allotment: Student S001 -> Room 101 Block A  
[PASS] WB10 pref=1 -> exact match -> ALLOTTED: 101  
WB11: rankByMerit minDiff fallback loop (L114-120) - no exact match for pref=3  
[NOTIFY] Allotment: Student S001 -> Room 101 Block A  
[PASS] WB11 no exact match -> minDiff fallback -> ALLOTTED (nearest room)  
WB12: D5=TRUE (feeStatus=null) - null-safe equals check -> ALLOTMENT_CANCELLED  
[NOTIFY] Allotment: Student S001 -> Room 301 Block B  
[NOTIFY] Cancelled: Allotment for student S001 cancelled due to non-payment.  
[PASS] WB12 feeStatus=null -> ALLOTMENT CANCELLED + room freed  
WB13: D3=FALSE boundary (pref=1, min valid) - full allotment path  
[NOTIFY] Allotment: Student S001 -> Room 101 Block A  
[PASS] WB13 pref=1 (min valid) -> D3 FALSE -> ALLOTTED: 101  
WB14: D3=FALSE boundary (pref=5, max valid) - full allotment path  
[NOTIFY] Allotment: Student S001 -> Room 501 Block C  
[PASS] WB14 pref=5 (max valid) -> D3 FALSE -> ALLOTTED: 501  
WB15: [P8] D4 both branches - S001 allotted, S002 waitlisted (1 room only)  
[NOTIFY] Allotment: Student S001 -> Room 301 Block B  
[NOTIFY] Waitlist: Student S002 added to waitlist.  
[PASS] WB15 S001 -> ALLOTTED, S002 -> WAITLISTED (D4 both branches)
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RESULTS: PASSED=15 FAILED=0 TOTAL=15  
PASS RATE: 100%
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